# A HYBRID ARCHITECTURE FOR 2D GAME DEVELOPMENT: THE CASE OF ASTEROID MINER


## Authors

Luis Alejandro Morales, Juan Manuel Otálora,
Julian Felipe Donoso Hurtado

**Universidad Distrital Francisco Jose De Caldas**


## Introduction

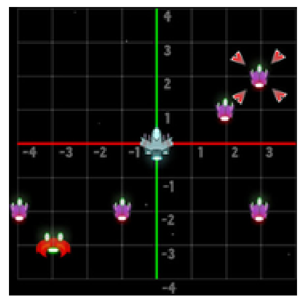
Modern video games often require efficient decision-making in resource-constrained environments, especially in scenarios involving resource management and navigation. This project explores how classical algorithms can be applied to optimize these processes in a 2D game environment, providing both real-time performance and analytical insights.

## Objective

To design a 2D game that maximizes resource collection under fuel constraints by applying and comparing different algorithmic strategies, including greedy and backtracking approaches.

## Methodology

### Background



The development of efficient decision-making systems is a fundamental challenge in modern video game design, particularly in resource management. Asteroid Miner is a 2D environment where a mining unit must navigate a field of coordinates to collect resources under strict fuel constraints.

### Algorithmic Logic

**Greedy Algorithm**

Used for real-time asteroid selection. It prioritizes targets based on the Value-to-Distance ratio, offering high speed and low computational cost for immediate decisions.

**Backtracking Model**

Selected for optimal route planning. It systematically explores possible sequences of asteroid visits to ensure the path maximizes total resources without exceeding the fuel limit.

## Game

The system is structured as a progressive 2D game. Each level challenges the player with increasing difficulty metrics via asteroid count and tighter fuel margins. At the end of each level, performance is quantified through an explicit evaluation matrix:

- 3 Stars (⋆⋆⋆): Collected value equals the mathematically optimal solution.
- 2 Stars (⋆⋆): At least 75% of the optimal value collected.
- 1 Star (⋆): Minimum resource threshold met.
- 0 Stars (X): Failed to return to base or fell below the minimum resource thres

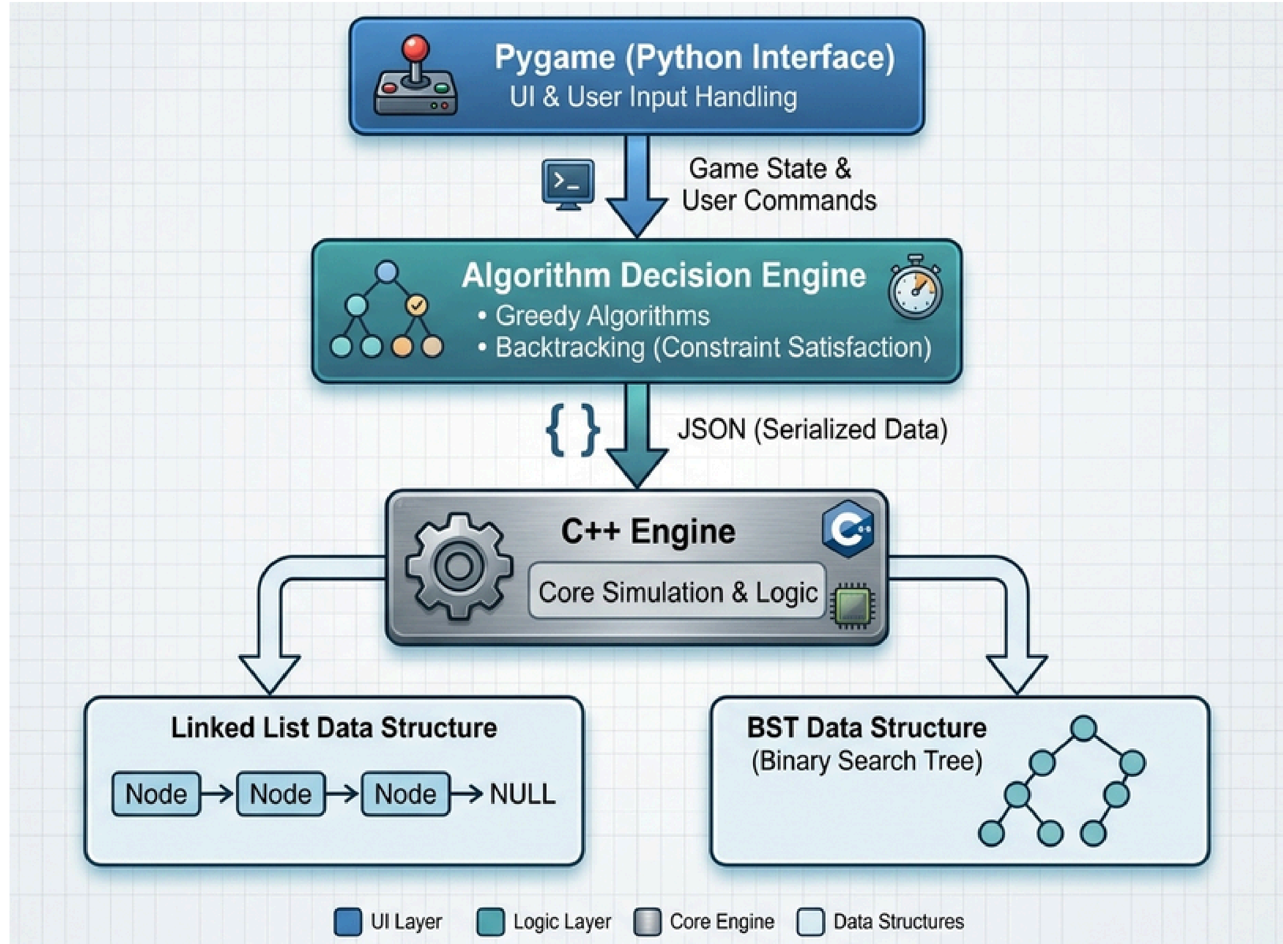

## Results

The preliminary results of the system demonstrate clear differences between the implemented algorithmic approaches in terms of efficiency and optimization. The greedy algorithm, based on the value-distance relationship, provides fast and computationally efficient solutions; however, tbut does not always achieve the maximum possible resource collection.

On the other hand, the backtracking algorithm guarantees the optimal solution by exhaustively exploring all feasible routes under the fuel constraint. This ensures maximum resource collection and a safe return to base. However, this approach has a significantly higher computational cost.

Although they function differently and yield distinct results, we believe that both algorithms will be very useful for the development and efficiency of our game.

## Architect




## Conclusion

This project successfully showed how classical algorithmic techniques can be brought to life inside an interactive game. By combining greedy heuristics with backtracking, the system offered both fast real-time recommendations and guaranteed optimal routes under fuel constraints. The use of linked lists and binary search trees in C++ effectively met the requirements, delivering efficient data organization and reliable persistent storage.The hybrid Python–C++ architecture worked smoothly, keeping gameplay logic cleanly separated from data management. Simple JSON communication handled the interaction between both languages without unnecessary complexity.Future work could focus on balancing the game with larger maps, implementing self-balancing trees like AVL, adding dynamic asteroid generation, and exploring more advanced optimization methods such as branch-and-bound or genetic algorithms.